MATH / BC GRADE 10

Slope & linear relations

Connect a table, a graph, and the equation of a line.

Foundations of Mathematics and Pre-calculus 10

01 Slope is a rate of change

For two distinct points, slope m = (y2 - y1) / (x2 - x1), when x2 is not x1. Use the same order for both differences. Include units when a graph models a real situation.

02 Read y = mx + b

m is the slope. b is the y-intercept: the y-value when x = 0. A horizontal line has slope 0; a vertical line has undefined slope and cannot be written in this form.

03 Use a point to find b

Once you know m, substitute one known x,y pair into y = mx + b. Solve for b, then verify the other point.

Worked example

Find the line through (1, 5) and (3, 9)

m = (9 - 5)/(3 - 1) = 2. Substitute (1,5): 5 = 2(1) + b, so b = 3. The equation is y = 2x + 3. Check (3,9): 2(3) + 3 = 9.

Watch out: A y-intercept is not any point on the graph: its x-coordinate must be 0.

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Try it. Then check it.

Work through these questions before looking at the answers.

1. Find the slope through (2,7) and (6,15).

2. For y = -3x + 8, state the slope and y-intercept.

3. A service costs $5 plus $2 per visit. Write the cost C for v visits and find C when v = 4.

Answer key

1. m = (15 - 7)/(6 - 2) = 8/4 = 2.

2. Slope: -3. The y-intercept is (0,8).

3. C = 2v + 5. For four visits, C = 2(4) + 5 = $13.

One-minute recap

Close the notes. Explain one key idea in your own words, then retry the question you found most difficult.

About this guide

Original Study Bridge topic review prepared with AI assistance. Suggested course fit, not complete or official course coverage. Compare with your teacher's materials. Examples and practice questions are newly written.

Official course context: Foundations of Mathematics and Pre-calculus 10

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